# Chapter 7: Behavioral Strategies

ECON 5001
Prof. Paul J. Healy
Based on: *Game Theory* by Giacomo Bonanno

Updated 2026-11-10 at 01:22:40

# Where We Left Off

Chapter 4: game trees with **information sets**, solved by SPNE

- Solve the smallest subgames first, work backwards
  - A generalization of backwards induction
- We said if a subgame has **no** NE, then there's **no SPNE**

Chapter 6: **mixed strategies**

- **Nash's Theorem**: every finite game (or subgame) has a (possibly mixed) NE
- So the "no NE in a subgame" problem can be fixed up
- But we have to allow mixing inside a game tree

**Chapter 7:**

1. Put lotteries and mixing into game trees
2. Find a **simpler** way to mix in trees: behavioral strategies
3. Show that a SPNE always exists
4. See what SPNE still gets wrong

# Lotteries in Game Trees

# “The Game” 2013

2013 OSU @ Michigan. Urban Meyer vs. Brady Hoke. 1,129 total offensive yards
Was 42-35. With 32 sec left, MI scores TD. Now 42-41.

- Brady Hoke: **Kick** to tie/OT or **Go for 2** to win

- **Kick**: Good 95% of the time, almost surely leads to overtime
  - Overtime: let’s say it’s 50-50 who wins
- **Go for 2**: Model it as a $2 \times 2$ subgame
  - Michigan: **Run** or **Pass**
  - OSU: **Stop Run (SR)** or **Stop Pass (SP)**

Outcome Payoffs: $u(\text{Win}) = 100$, $u(\text{Lose}) = 0$ for both teams

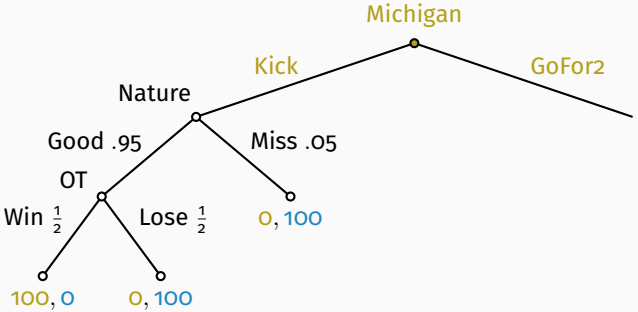
How to draw this tree?

# Drawing the Tree



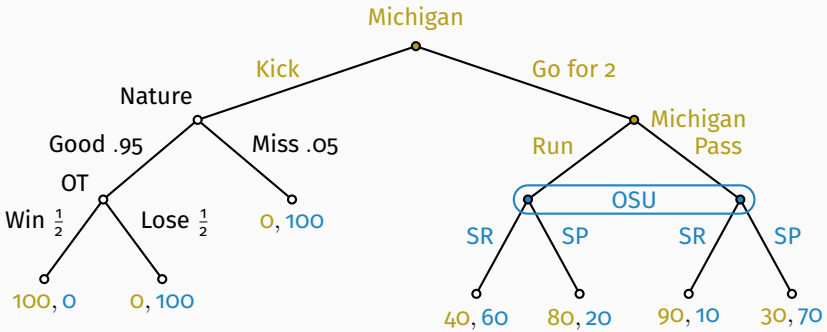

Michigan moves first

# Drawing the Tree



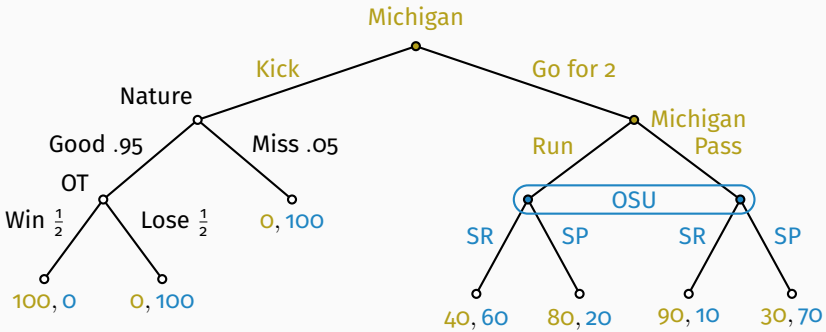

**Kick:** two moves by “Nature”. Each branch has a probability

# Drawing the Tree




**GoFor2:** a simultaneous-move game hanging off the tree

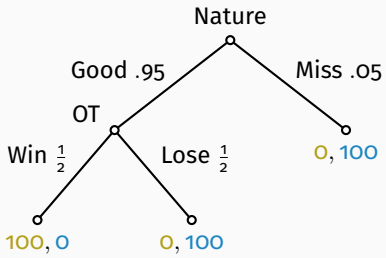
# Drawing the Tree




How many proper subgames? (Include moves by Nature.) _____

# Nature Gives a Lottery Over Outcomes

After Michigan chooses Kick:




Overtime gives Michigan $\begin{pmatrix} \text{Win} & \text{Lose} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$U(\text{OT}) =$ _____

Kick gives Michigan $\begin{pmatrix} \text{OT} & \text{Lose} \\ .95 & .05 \end{pmatrix}$

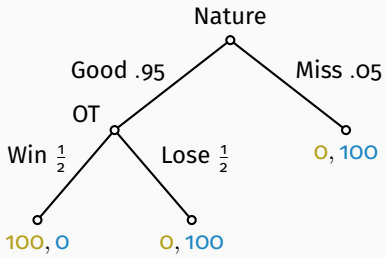
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**Rule:** Same as Chapter 6. Use expected utilities

Bonanno (§7.1) writes the lottery as the payoff instead of drawing Nature. Using Nature is more standard.

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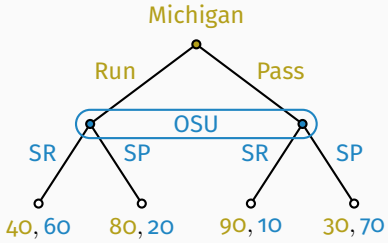
**Rule:** Same as Chapter 6. Use expected utilities

So: replace the entire Kick branch with _____

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# Subgame-Perfect Equilibrium: Allowing Mixing

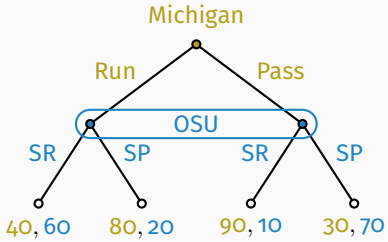
# Step 1: Solve the Subgame After “Go for 2”




| Michigan \ OSU | Stop Run | Stop Pass |
|---|---|---|
| Run | 40, 60 | 80, 20 |
| Pass | 90, 10 | 30, 70 |

Payoffs = Expected utility = Prob. of Win

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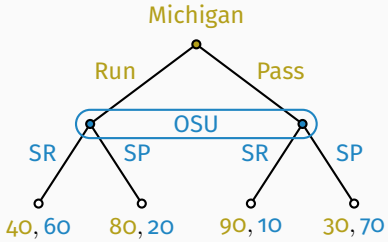



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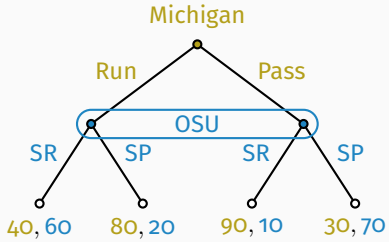



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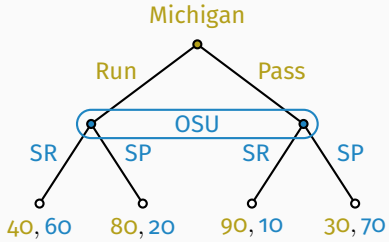



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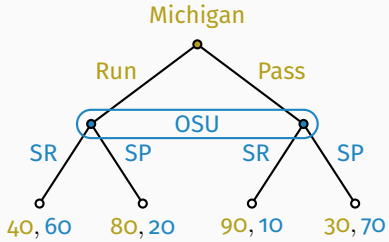



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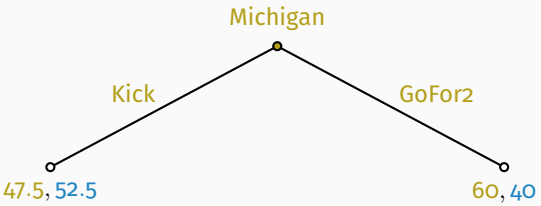

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Expected Payoffs in MSNE:
Michigan: 60 (percent chance of winning)
OSU: 40 (percent chance of winning)

# Step 2: Replace and Solve




**Kick:** replace that subgame by its expected utility
Go for 2: the subgame, replaced by its **mixed** NE payoffs

Michigan chooses: _____

# What Actually Happened in 2013



Photo: Shelby Lum, *The Lantern*

**The Decision:**
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Consistent with SPNE!

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**Did Hoke make a bad choice?**

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Hoke went for 2.
Consistent with SPNE!

**The Play:**
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**Did Hoke make a bad choice?**
A 60% chance still
loses 40% of the time

# The SPNE

The SPNE is fully specified by giving the probability of every action at **every** decision node:

$$\text{Michigan: } \sigma_1^* = \left( \begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ \underline{\quad} & \underline{\quad} & \underline{\quad} & \underline{\quad} \end{array} \right) \qquad \text{OSU: } \sigma_2^* = \left( \begin{array}{cc} \text{SR} & \text{SP} \\ \underline{\quad} & \underline{\quad} \end{array} \right)$$

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That's called a **behavioral strategy**

# Selten's Theorem

**Theorem (Selten, 1965)**
Every finite extensive-form game with cardinal payoffs has **at least one** SPNE (possibly involving mixed strategies).

(**Finite game:** Finite # players, finite # of actions, tree doesn't go forever.)

Why?

- Every subgame is a finite game
- Nash: every finite game has a (mixed) NE
- So the SPNE algorithm can give a prediction at every subgame
- And the game is finite, so the algorithm doesn't run forever

Reinhard Selten shared the 1994 Nobel Prize with John Nash and John Harsanyi.

# Behavioral Strategies

# Michigan's Strategies

A strategy: one choice at every information set (Chapter 4)

Michigan has two information sets, two choices each:

$$S_{MI} = \{\ (\text{Kick}, \text{Run}),\ (\text{Kick}, \text{Pass}),\ (\text{GoFor2}, \text{Run}),\ (\text{GoFor2}, \text{Pass})\ \}$$

- (Kick, Run) and (Kick, Pass) are different strategies, even though Kick precludes the Run/Pass decision.

Mixed strategy (Chapter 6): a probability for each of the 4 strategies

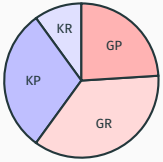
$$\begin{pmatrix} (\text{Kick}, \text{Run}) & (\text{Kick}, \text{Pass}) & (\text{GoFor2}, \text{Run}) & (\text{GoFor2}, \text{Pass}) \\ p & q & r & 1-p-q-r \end{pmatrix}$$

- Needs _____ numbers

# Two Ways To Randomize

**Mixed strategy**

- Before the game: spin one big wheel over **complete plans**
- Then follow the plan




**Behavioral strategy**

- At each information set, when you get there: flip that set's coin
- One coin per information set




Behavioral strategy: $\left(\begin{array}{cc|cc} \text{Kick} & \text{GoFor2} & \text{Run} & \text{Pass} \\ p & 1-p & q & 1-q \end{array}\right)$ Needs _____ numbers

# Formal Definition: Behavioral Strategy

**Definition (Bonanno 7.1.1)**
A **behavioral strategy** for a player is a list of probability distributions, one for every information set of that player. These give the probability the player plays each action at each information set.

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**Fact:** If a player has only one information set then mixed $=$ behavioral (like OSU)

# Are They the Same?

Suppose Michigan's coach uses this mixed strategy:

$$\sigma_1 = \begin{pmatrix} (\text{Kick}, \text{Run}) & (\text{Kick}, \text{Pass}) & (\text{GoFor2}, \text{Run}) & (\text{GoFor2}, \text{Pass}) \\ 0.1 & 0.3 & 0.36 & 0.24 \end{pmatrix}$$

Find a behavioral strategy that does the same thing:

$\Pr(\text{Kick}) =$ _____ $\Pr(\text{GoFor2}) =$ _____

$\Pr(\text{Run} \mid \text{GoFor2}) =$ _____ $\Pr(\text{Pass} \mid \text{GoFor2}) =$ _____

**Notation**: $\Pr(\text{A} \mid \text{B})$ means "Probability of A, given *B* already occurred."
This is called a *conditional probability*.

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Check: $\Pr(\text{GoFor2 and Run}) =$ _____ $= 0.36$ ✓

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Check: $\Pr(\text{GoFor2 and Run}) =$ _____ $= 0.36$ ✓

**Lesson:** The 0.1 vs. 0.3 split never matters. The behavioral strategy doesn't require it.

# Kuhn's Theorem

**Theorem (Kuhn, 1953)**
In games with **perfect recall**, for every mixed strategy there is a behavioral strategy that gives the same probability over terminal nodes

Recall perfect recall (Chapter 4):

- Never forget what you knew
- Never forget what you did

**Takeaway:** Use behavioral strategies. They're simpler, and nothing is lost

**Notation:** We'll use $\sigma_i$ to describe either a mixed or behavioral strategy

Bonanno works through his own example (Figure 7.3) on pp. 239–241.

# Two Ways To Be Unpredictable

Randomize **as you go** (behavioral):

- **Play-calling**: each down, run or pass?
- **Pitching**: each pitch, fastball or curveball?
- **Poker**: each betting round, bluff or fold?
- **Penalty kicks**: each kick, left or right?

Randomize **a whole plan in advance** (mixed):

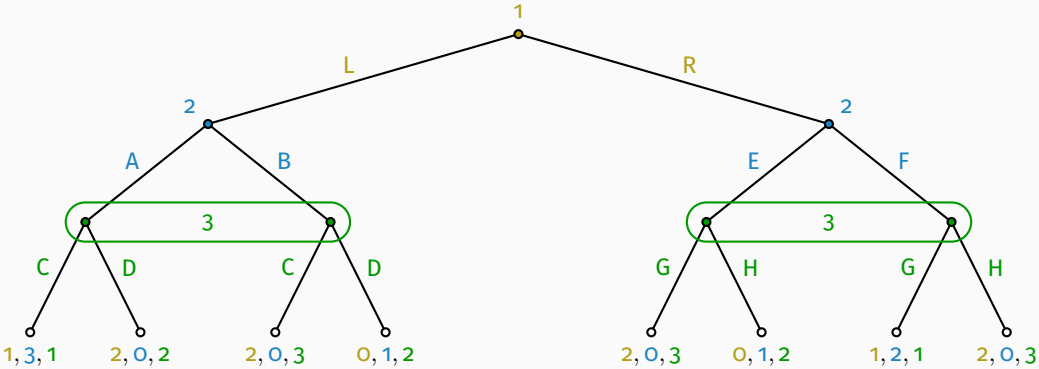
- **Security**: LAX's ARMOR software draws each day's checkpoint schedule
- **Inspections**: random drug-test lists, tax audits, restaurant inspections
- **Exams**: a random version of the test, handed out before it starts

Kuhn: with perfect recall, the two are equally powerful

# A Bigger Example

# Figure 7.7

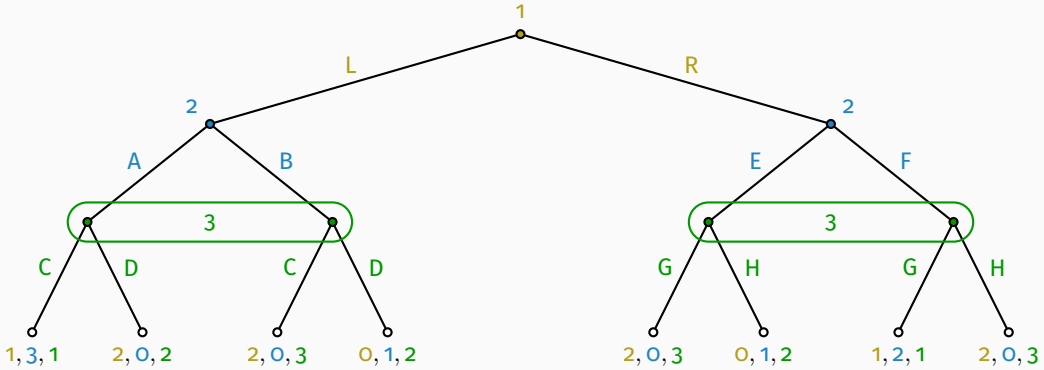
Find the SPNE




How many proper subgames? _____ Any pure NE in them?

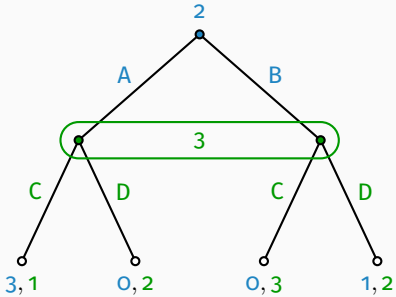
# Figure 7.7

Find the SPNE




How many proper subgames? 2 Any pure NE in them? No!

# Figure 7.8: The Left Subgame




Player 3

| Player 2 | C | D |
|---|---|---|
| A | 3, 1 | 0, 2 |
| B | 0, 3 | 1, 2 |

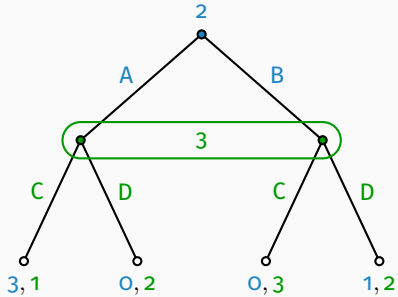
$q = \Pr(C)$ makes P2 indifferent:

$p = \Pr(A)$ makes P3 indifferent:

Payoffs (P1, P2, P3): (_____, _____, _____)

P1's payoffs come from the tree

# Figure 7.8: The Left Subgame




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$q = \Pr(C)$ makes P2 indifferent:

$3q = 1 - q \;\Rightarrow\; q = \frac{1}{4}$

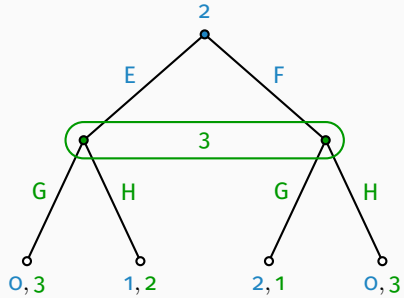
$p = \Pr(A)$ makes P3 indifferent:

$p + 3(1-p) = 2 \;\Rightarrow\; p = \frac{1}{2}$

Payoffs (P1, P2, P3): $\left(\ \frac{9}{8}\ ,\ \frac{3}{4}\ ,\ 2\ \right)$

P1's payoffs come from the tree

# Figure 7.10: The Right Subgame



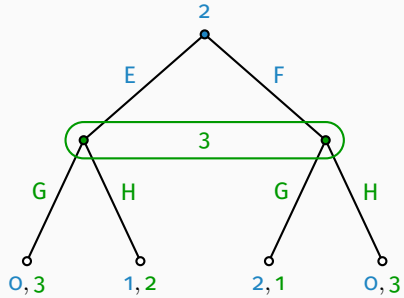

| Player 2 \ Player 3 | G | H |
|---|---|---|
| E | 0, 3 | 1, 2 |
| F | 2, 1 | 0, 3 |

$q = \Pr(G)$ makes P2 indifferent:

$p = \Pr(E)$ makes P3 indifferent:

Payoffs (P1, P2, P3): (_____, _____, _____)

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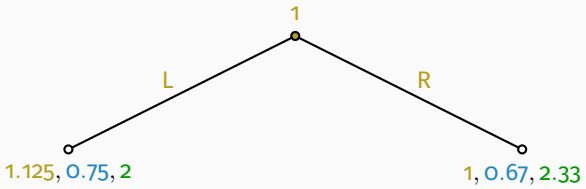
$1 - q = 2q \;\Rightarrow\; q = \frac{1}{3}$

$p = \Pr(E)$ makes P3 indifferent:

$3p + (1-p) = 2p + 3(1-p) \;\Rightarrow\; p = \frac{2}{3}$

Payoffs (P1, P2, P3): $\left(\ 1\ ,\ \frac{2}{3}\ ,\ \frac{7}{3}\ \right)$

# Figure 7.11: What's Left



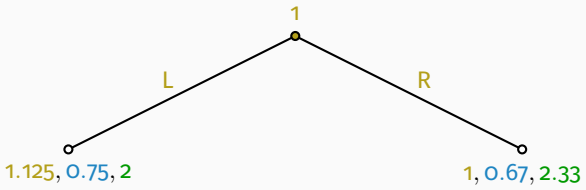

Player 1 chooses: _____

**SPNE**, in behavioral strategies:

$$\text{P1:} \begin{pmatrix} L & R \\ ___ & ___ \end{pmatrix} \qquad \text{P2:} \left(\begin{array}{cc|cc} A & B & E & F \\ ___ & ___ & ___ & ___ \end{array}\right)$$

$$\text{P3:} \left(\begin{array}{cc|cc} C & D & G & H \\ ___ & ___ & ___ & ___ \end{array}\right)$$

# Figure 7.11: What's Left




Player 1 chooses: $L$

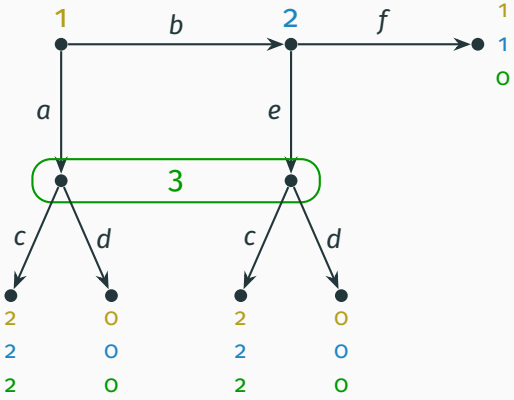
**SPNE**, in behavioral strategies:

$$\text{P1:} \begin{pmatrix} L & R \\ 1 & 0 \end{pmatrix} \qquad \text{P2:} \left(\begin{array}{cc|cc} A & B & E & F \\ \frac{1}{2} & \frac{1}{2} & \frac{2}{3} & \frac{1}{3} \end{array}\right)$$

$$\text{P3:} \left(\begin{array}{cc|cc} C & D & G & H \\ \frac{1}{4} & \frac{3}{4} & \frac{1}{3} & \frac{2}{3} \end{array}\right)$$

# A Critique of SPNE
# Due to Selten Himself

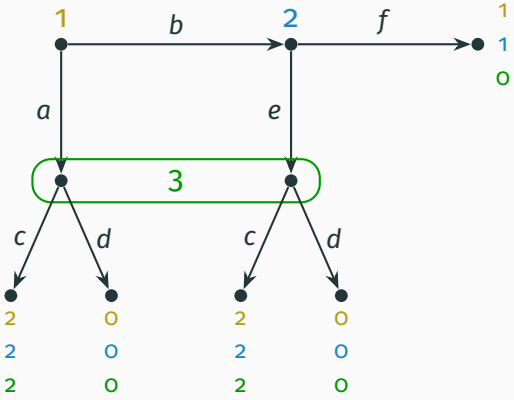
# A Critique of SPNE: Selten's Horse




Proper subgames? _____

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

# A Critique of SPNE: Selten's Horse

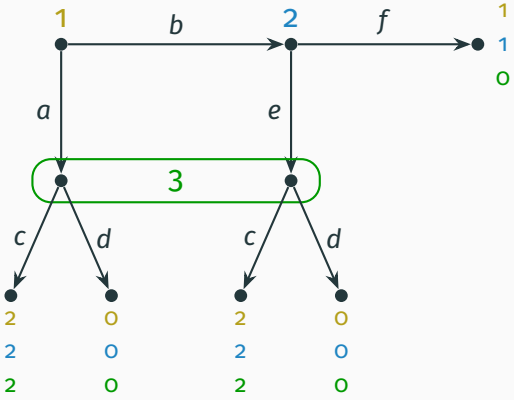



Proper subgames? _____

So SPNE = NE

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

# A Critique of SPNE: Selten's Horse



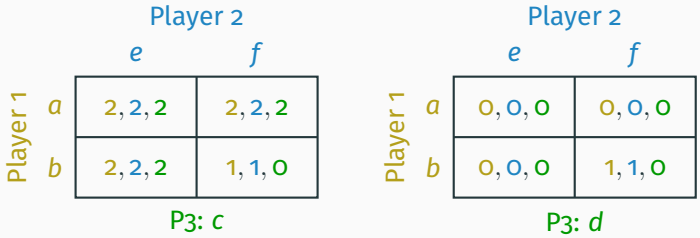

Proper subgames? _____

So SPNE = NE

To find the NE: write the strategic form

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

# Selten's Horse: Strategic Form




P1 picks the row, P2 the column, P3 the matrix

# Selten's Horse: Strategic Form

P3: *c*

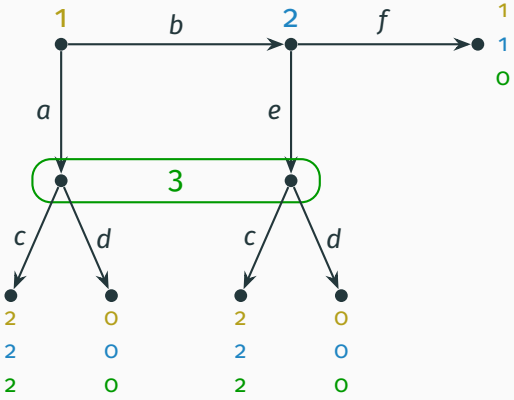
| Player 1 \ Player 2 | *e* | *f* |
|---|---|---|
| *a* | 2, 2, 2 | 2, 2, 2 |
| *b* | 2, 2, 2 | 1, 1, 0 |

P3: *d*

| Player 1 \ Player 2 | *e* | *f* |
|---|---|---|
| *a* | 0, 0, 0 | 0, 0, 0 |
| *b* | 0, 0, 0 | 1, 1, 0 |

Pure NE: _____ _____ _____ _____

# Selten's Horse




Pure NE (all are SPNE):

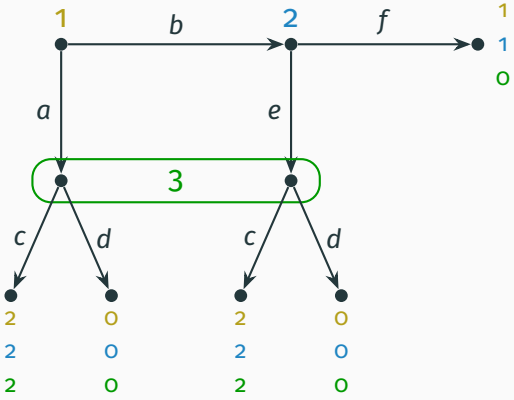
$(a, e, c)$ $(a, f, c)$

$(b, e, c)$ $(b, f, d)$

Are they all reasonable???

Bonanno Figure 7.12. Named by Selten (1975) for its shape.

# Problem: the $(b, f, d)$ SPNE

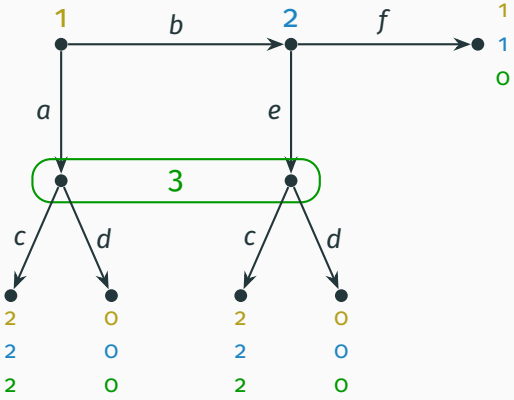



Consider the $(b, f, d)$ SPNE

P1 plays $b$, P2 plays $f$. P3 never moves

Bonanno typo, p. 249: for $(b, f, d)$ it says "Player 1 plays $a$". Should be $b$.

# Problem: the $(b, f, d)$ SPNE




Consider the $(b, f, d)$ SPNE

P1 plays $b$, P2 plays $f$. P3 never moves

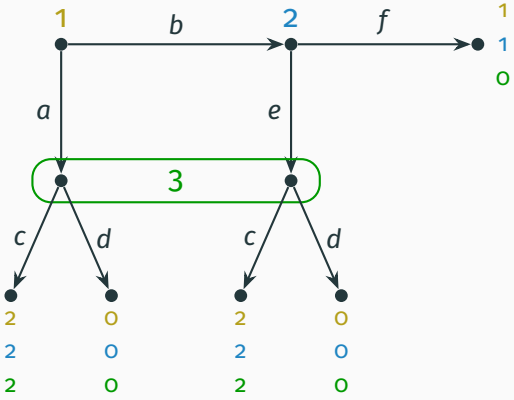
But **if** P3 had to move?

Left node: $c$ _____ vs. $d$ _____

Right node: $c$ _____ vs. $d$ _____

Bonanno typo, p. 249: for $(b, f, d)$ it says "Player 1 plays $a$". Should be $b$.

# Problem: the $(b, f, d)$ SPNE




Consider the $(b, f, d)$ SPNE

P1 plays $b$, P2 plays $f$. P3 never moves

But **if** P3 had to move?

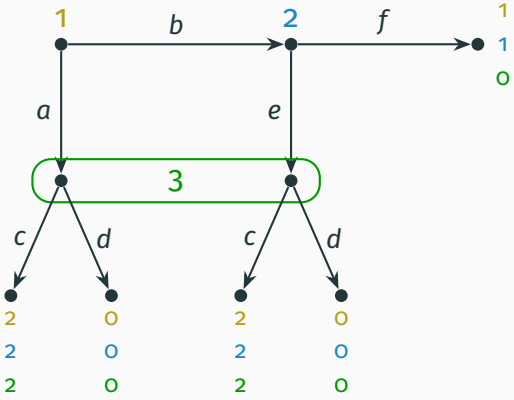
Left node: $c$ _____ vs. $d$ _____

Right node: $c$ _____ vs. $d$ _____

$d$ is **strictly dominated** at P3's infoset!

Bonanno typo, p. 249: for $(b, f, d)$ it says "Player 1 plays $a$". Should be $b$.

# Problem: the $(b, f, d)$ SPNE




Consider the $(b, f, d)$ SPNE

P1 plays $b$, P2 plays $f$. P3 never moves

But **if** P3 had to move?

Left node: $c$ ____ vs. $d$ ____

Right node: $c$ ____ vs. $d$ ____

$d$ is **strictly dominated** at P3's infoset!

If 2 knows 3 will play $d$, 2 switches to $e$

**Problem:** Player 3's info set isn't the root of a subgame. In this SPNE it's never reached, so anything is a best reply

Bonanno typo, p. 249: for $(b, f, d)$ it says "Player 1 plays $a$". Should be $b$.

# So What's Missing?

SPNE checks for rationality only in **subgames**

- A subgame can't cut an information set
- Maybe that's not strict enough?

The fix: Check that strategies are rational at **every** infoset (not just subgames)

**Goal:** $\{\text{better solution concept}\} \subseteq SPNE \subseteq NE$

**Coming in Chapter 11:** a new solution concept that fixes this

- **Weak sequential equilibrium**:
  1. Players have beliefs over their infosets (*e.g.*, Pr(right node)=1)
  2. Require players best respond to their beliefs at every infoset (*e.g.*, play $c$)

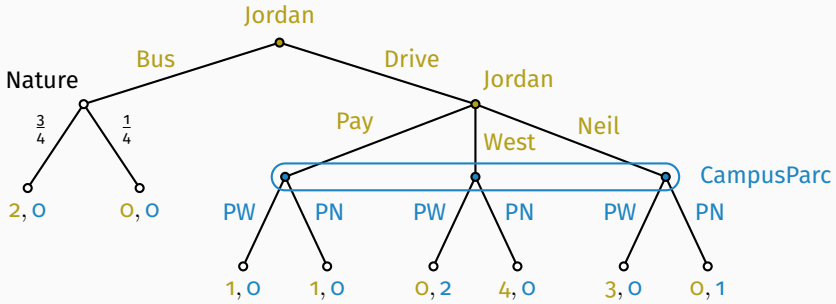
Another fix: **trembling-hand perfect equilibrium** (Selten, 1975). Not covered in this class.

# Let’s Work One Out

# Gameday Parking, Part 2

Jordan first decides how to get to the game:

- **Bus** (COTA): on time with chance $\frac{3}{4}$ (payoff 2), else late (payoff 0)
- **Drive**: then Chapter 6's parking game. CampusParc can't see where Jordan parked



# Gameday Parking: The Drive Subgame

| Jordan \ CampusParc | Patrol West | Patrol Neil |
|---|---|---|
| Pay | 1, 0 | 1, 0 |
| West | 0, 2 | 4, 0 |
| Neil | 3, 0 | 0, 1 |

The subgame after Drive. Same as Chapter 6's parking game

# Gameday Parking: The Drive Subgame

| Jordan \ CampusParc | Patrol West | Patrol Neil |
|---|---|---|
| Pay | 1, 0 | 1, 0 |
| West | 0, 2 | 4, 0 |
| Neil | 3, 0 | 0, 1 |

No cell is fully underlined: no pure NE. Need the mixed NE (Chapter 6)

# Gameday Parking, Part 2: Answers

Jordan's pure strategies: _____ Numbers for a mixed strategy: _____ Behavioral: _____

Expected payoff from Bus: _____

Subgame NE: Jordan $\begin{pmatrix} \text{Pay} & \text{West} & \text{Neil} \\ ____ & ____ & ____ \end{pmatrix}$ CampusParc $\begin{pmatrix} \text{PW} & \text{PN} \\ ____ & ____ \end{pmatrix}$

Subgame payoffs: (_____, _____) Jordan chooses: _____

SPNE: Jordan $\left(\begin{array}{cc|ccc} \text{Bus} & \text{Drive} & \text{Pay} & \text{West} & \text{Neil} \\ ____ & ____ & ____ & ____ & ____ \end{array}\right)$ CampusParc $\begin{pmatrix} \text{PW} & \text{PN} \\ ____ & ____ \end{pmatrix}$

Bus is worth it if it's on time with chance at least _____

# Gameday Parking, Part 2: Answers

Jordan's pure strategies: 6 Numbers for a mixed strategy: 5 Behavioral: 3

Expected payoff from Bus: $\frac{3}{2}$

Subgame NE: Jordan $\begin{pmatrix} \text{Pay} & \text{West} & \text{Neil} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{pmatrix}$ CampusParc $\begin{pmatrix} \text{PW} & \text{PN} \\ \frac{4}{7} & \frac{3}{7} \end{pmatrix}$

Subgame payoffs: ( $\frac{12}{7}$ , $\frac{2}{3}$ ) Jordan chooses: Drive

SPNE: Jordan $\left(\begin{array}{cc|ccc} \text{Bus} & \text{Drive} & \text{Pay} & \text{West} & \text{Neil} \\ 0 & 1 & 0 & \frac{1}{3} & \frac{2}{3} \end{array}\right)$ CampusParc $\begin{pmatrix} \text{PW} & \text{PN} \\ \frac{4}{7} & \frac{3}{7} \end{pmatrix}$

Bus is worth it if it's on time with chance at least $\frac{6}{7}$

# What Have We Learned, Charlie Brown?

- Nature in a tree? Replace the lottery with its **expected utility**
- **Behavioral strategy**: one probability distribution per information set
  - Mixed: spin one wheel over whole plans. Behavioral: flip as you go
- **Kuhn (1953)**: with perfect recall, behavioral $=$ mixed
- **Selten (1965)**: with cardinal payoffs, a SPNE always exists
  - Solve subgames with mixed NE when there's no pure one
- SPNE can still allow silly choices at unreached infosets (Selten's horse)
- The fix comes in Chapter 11: weak sequential equilibrium

# Reading & Practice

- Bonanno, *Game Theory*, Chapter 7
  - §7.1: read for behavioral strategies and Kuhn's theorem. Skim the formal "game with cardinal payoffs" part
  - §7.2–7.3: read carefully
- Exercises: §7.4.1 (7.1, 7.2, 7.3), §7.4.2 (7.5, 7.7)
  - 7.4 and 7.6 review Chapter 5's utility calibration
  - 7.8 is harder. 7.9 (the mugger) is a challenge
- Solutions in §7.5. Try them first